

Revision Notes for Version 5

A Cauchy Limit for a Sawtooth Sum

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These notes summarize the changes from the initial manuscript to version 5. For reading, the v5 PDF should be used as the main text; the line references below refer to the accompanying v5 TeX source. The revision to Proposition 6.4 is also described separately in a short note.

Front matter and introduction

Author information

I added the author name, Sangyoon Kwon, to the front matter.

v5 line 31

AI-use statement

I moved the short front-page disclosure to the end matter and replaced it with a fuller AI-use statement describing the division between human direction, AI assistance, and my responsibility for the final arguments and text.

v5 lines 1572-1580

Sign convention and context

I added the opposite-sign convention and the measure-preserving change $\alpha \rightarrow 1-\alpha$. I also added a comparison with Vardi's limit theorem for Dedekind sums.

v5 lines 53-68

Sections 1 and 2

Principal split

I replaced braces around $W(u_j)-W(\theta_j)$ with parentheses so that they cannot be read as a set or a fractional-part operation.

v5 lines 233-238

Levy constant

I corrected the description of λ : it is the Levy constant, while the Gauss-map Lyapunov exponent is 2λ .

v5 lines 248-257

Section 3

Lemma 3.2 and Lemma 3.1(i)

I corrected the density citation and wrote the exact cylinder-transfer identity before the use of Lemma 3.1(i). I retained Lemma 3.1(i) because this identity rewrites the relevant conditional expectation but does not by itself provide the exponential $O(\rho^M)$ estimate required in Lemma 3.2. The same estimate is used in the zero-mode factorizations in Propositions 4.1 and 4.2. Lemma 3.3 no longer uses this input.

v5 lines 360-374

Lemma 3.3 and display (21)

I replaced the natural-extension argument with a proof by word reversal and continuant symmetry. The resulting estimate is $4\eta + 4F_{j+1}^{-2}$, uniformly for every nonzero integer pair (r,s) and every $\eta > 0$.

v5 lines 397-442

Lemma 3.4 statement

I made the converse phase-freeze clause explicit by requiring $Q \in \mathbb{Z} \setminus \{0\}$.

v5 lines 458-466

Lemma 3.4 notation

I renamed the continuation denominators to (ℓ, ℓ') and the prefix denominators to (q, q_-) to avoid using the same symbols for three different quantities.

v5 lines 467-484

Section 4

Proposition 4.1: zero mode

I now derive $\text{Var}(g_l) \leq 2 \sum_a |c_l(a, 0)| \leq 2L^D$ from the coefficient l_1 bound in display (24). Counting jumps alone would only give the weaker $O(L^{(2D)})$ estimate.

v5 lines 521-535

Identity (31)

I extended the character calculation to the full window $-R-1 \leq t \leq R$ and specified the digit word used in the forward and backward substitutions.

v5 lines 603-623

Proposition 4.2: constants

I separated the previous generic exponent into c_{ld} , c_{osc} , and c_{mix} , and stated how these constants may depend on the fixed monomials.

v5 lines 640-650

Proposition 4.2: later mode

I introduced κ , δ , and γ with $\kappa + 3\delta < 80\lambda$, and separated the anti-concentration, denominator-deviation, and oscillatory error terms.

v5 lines 654-713

Proposition 4.2: earlier mode

I rewrote the pre- and post-resonance cases using explicit t_- , t_+ , γ , γ_+ , and c_{mix} , including the restoration of the discarded cylinders.

v5 lines 714-789

Section 5

Point process

I added the lead-in 'Define the marked point process' before display (34).

v5 line 801

Proposition 5.1

I declared the auxiliary exponent A_0 before it is used.

v5 line 821

Jackson approximation

I replaced the ambiguous A by A_0 , corrected the Zygmund reference to Vol. I, Chap. III, and used A_0 consistently in the approximation estimates.

v5 lines 840-852

Small-jump variance

I declared A_0 locally, made the epsilon- and L-uniform Lipschitz constant explicit, and used A_0 consistently in the covariance sum.

v5 lines 965-982

Section 6

Lemma 6.1

I replaced 'a nonzero integer sequence' with 'an integer sequence, not identically zero'.

v5 line 1050

Window laws

I named the stationary window law μ_R and introduced the consistent projections $\pi_{\{R', R\}}$, together with their pushforward and actual-window identities.

v5 lines 1153-1165

Lemma 6.3

I made the finite word support explicit, used uniform anti-concentration constants, separated κ from δ , and clarified the order of the digit and complete-quotient truncations.

v5 lines 1176-1245

Proposition 6.4

I introduced $B^\wedge(R)$ as a bounded almost-everywhere continuous window function and added the full approximation $B^\wedge(R) \rightarrow G \rightarrow G_M \rightarrow G_{M1_E} \rightarrow P_{\{R,M\}}$. I also corrected the lifted L2 norm in display (55) and made the subsequent use of Proposition 4.2 explicit.

v5 lines 1305-1463

Section 7

Lemma 7.1: end terms

I replaced the one-direction wording about omitted indices with the symmetric difference I_n triangle J_n , treated both endpoint directions, and added the Markov inequality step.

v5 lines 1506-1530

End matter and references

Acknowledgments, disclosure, and references

I added acknowledgments, rewrote the AI-use statement, added Vardi (1993), and expanded the Zygmund entry to Vols. I-II.

v5 lines 1557-1629

Files to use

The v5 PDF is the reading copy. The v5 TeX is included for exact line references and for the formalization work. The separate Proposition 6.4 note isolates the revision around display (55); the present document lists the remaining changes in manuscript order.